Referee's Report on:

A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex-Pseudoconvex Constraints

—Nguyen Van The, Tran Ngoc Thang and Nguyen Hong Son

The authors study the nonlinear SFP with a normal box as the output set and a convex+pseudoconvex component mapping. By using Tchebyshev scalarization, the authors turn SFP into a closed-form max-type violation function. They propose a DTAP method. The stepsizes of DTAP are irrelevant to operator norm, Lipschitz, or Hölder modulus. They prove the strong convergence of the method and apply it to a feature-selection problem. The following are some comments on the current version of the paper.

Major issue:

1. Section 1 "Introduction" and Section 2 "Related work" are suggested to be reorganized. Both sections are suggested to be merged as Section 1. Furthermore, the Related work is unnecessary to be revisited in detail. After merging the Introduction and Related work, it is suggested to limit them to 4 pages. After all, the paper is a research paper, rather than an overview. Please shorten the paper as much as possible. Much of the content is redundant for a research paper. Otherwise, it is suggested that the authors submit it as an overview. Maybe 25 pages is enough for this paper by removing redundant content and shortening long sentences.

2. In order to make the paper more readable, the writing of the current version of the paper should be substantially improved. Many sentences are ambiguous. For me, I can NOT understand many long sentences in the text. Please describe the problem, algorithms, etc. in short sentences as much as possible. Please polish each sentence carefully. In the text, there are many nonsense descriptions; please make the paper concise and abridge it as much as possible.

3. The format of the references is inconsistent. Many important references are omitted for SFP methods. E.g., I remember the authors of [23], [35] also have some work on SFP with self-adaptive or decomposition methods.

4. Why do the authors present some numerical results in the Appendix?

Minor Issues:

1. page 2, line 18, what is the "reference point"? It seems to be ambiguous.

2. page 2, line 27, “differing densities” or “different densities”

3. page 2, line 35, please polish the sentence “The NSFP is not itself new”

4. page 2, line 39, “active index set” or “active set”?

5. page 2, lines 47-49, “We ask instead ... make precise” the sentence is difficult for me to understand.

6. page 3, lines 48-53, the sentences are too long; please use short sentences.

7. page 3, the authors present 5 contributions of this paper. Please summarize those main contributions instead of describing too many details.

8. page 5, “Rule (3) has since been extended”, please check if it is grammatically correct.

9. page 5, line 33, what is the meaning of “drives their proof is for free”? “the nonconvexity sits in the sets”?

10. page 5, line 35, what is “plain gradient inequality”?

11. page 5, line 44, what is “the closest predecessor”? Do you mean the seminal work?

12. page 5, line 48, “step” or “step size”?

13. page 7, line 31, dtap should be DTAP

14. page 7, line 39, “following the general framework of Combettes 10” should be “following the general framework of Combettes [10]”

15. page 9, the subtitles 3.1, 3.2 seem unnecessary. Subsection 3.3 is feature selection, which is suggested to be merged into the Numerical experiments section.

16. page 12, $H := \mathbb{R}^n$ has been stated in the Preliminaries section; why do we repeat $H := \mathbb{R}^n$ once again? Please write the paper as concisely as possible.

17. page 13, the subtitle of 4.3, what is the meaning of “standing assumption”?

18. page 18 line 40, $\sum\limits_{k=0}^{\infty} \left\| x_{k+1} - x_k \right\|^2 < \infty$ should be $\sum_{k=0}^{\infty} \|x_{k+1} - x_k\|^2 < \infty$. Please follow the advice of the AMS Math formula style as much as possible when presenting the formula.

19. page 20, line 36, “so no Young splitting is needed”. Please polish the writing as much as possible.

20. page 20, line 35, $K \cap \bigcap_{i \in I} {}^{cvx}$. I don’t think this expression is good.

I suggest the authors consider the above comments in their revision. Furthermore, it would be recommended if the authors could make the paper concise.